

76 Results

Arts, Sciences, and Engineering, Fall 2025, CS

Cart - Calendar View

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☐ Show Descriptions ☐ Show Sections

Enrollment Status: ● open ⛔ closed ⚠ waitlist

CS-0004	Teaching Computer Science
CS-0010	Programming Fundamentals
CS-0010	Programming Fundamentals
CS-0011	Introduction To Computer Science
CS-0011	Introduction To Computer Science
CS-0012	Cyber for Future Policymakers
CS-0013	How Systems Work
CS-0014	Emerging Scholars in Computer Science
CS-0015	Data Structures
CS-0015	Data Structures
CS-0020	Web Programming
CS-0023	Game Design
CS-0030	Programming for Data Science - Programming for Data Science
CS-0030	Programming for Data Science - Programming for Data Science
CS-0039	Ethics for AI, Robotics, and Human Robot Interaction
CS-0040	Machine Structure & Assembly-language Programming
CS-0045	Computer Organization
CS-0061	Discrete Mathematics
CS-0062	Discrete Math Skills Workshop
CS-0086	Object-oriented Programming For Graphical User Interfaces
CS-0093	Directed Study
CS-0097	Senior Capstone Project I
CS-0099	Internship Computer Science
CS-0105	Programming Languages

Principles and application of computer programming languages. Emphasizes ideas and techniques most relevant to practitioners, but includes foundations crucial for intellectual rigor: abstract syntax, lambda calculus, type systems, dynamic semantics. Case studies, reinforced by programming exercises. Grounding sufficient to read professional literature. CS 80 and CS 105 cannot both be taken for credit.

LECTURE

Select 1 Lecture, 1 Recitation

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80366	Regular	Mo, We 10:30AM - 11:45AM TBA Medford/Somerville	Richard Townsend	4	●	○

Meeting Dates 09/02/2025 - 12/08/2025

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Class Attributes		BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing					
Career		AS&E Undergrad/Grad					
Textbook		Textbook info					
Grading		Graded					
Enrollment Reqs							
Restricted to CS Majors, CS Minors, and CS Graduate Students. Undergraduates must have grades of C- or better in (CS15 and one of the following courses: CS 61, Math 61, or Math 65).							
Repeat for Credit		No					
Allow Multiple Enroll in Term		No					
Total Units Allowed		4					
Total Completions Allowed		1					
Instruction Mode		In Person					
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		120	0	120			
Wait List		20	0	20			

RECITATION

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
RA-RCT Details	80012	Regular	Th 10:30AM - 11:45AM TBA Medford/Somerville	Richard Townsend	0		
RB-RCT Details	80013	Regular	Th 12:00PM - 1:15PM TBA Medford/Somerville	Richard Townsend	0		
RC-RCT Details	80014	Regular	Th 1:30PM - 2:45PM TBA Medford/Somerville	Richard Townsend	0		
RD-RCT Details	80017	Regular	Th 3:00PM - 4:15PM TBA Medford/Somerville	Richard Townsend	0		
RE-RCT Details	80015	Regular	Th 4:30PM - 5:45PM TBA Medford/Somerville	Richard Townsend	0		
RF-RCT Details	80018	Regular	Th 6:00PM - 7:15PM TBA Medford/Somerville	Richard Townsend	0		
RG-RCT Details	80019	Regular	Th 7:30PM - 8:45PM TBA Medford/Somerville	Richard Townsend	0		
RH-RCT Details	80020	Regular	Fr 10:30AM - 11:45AM TBA Medford/Somerville	Richard Townsend	0		
RI-RCT Details	80039	Regular	Fr 12:00PM - 1:15PM TBA Medford/Somerville	Richard Townsend	0		
RJ-RCT Details	80040	Regular	Fr 1:30PM - 2:45PM TBA Medford/Somerville	Richard Townsend	0		

CS-0112 **Networks**

Computer Networks and Protocols. Design and implementation of computer communication networks, protocols, and applications, with an emphasis on the Internet protocol suite. Network architectures and programming interfaces. Data link, transport, and routing protocols. Congestion sources and remedies. Addressing and naming in local area and wide area networks. Network security and network management.

LECTURE

LECTURE							
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80418	Regular	Tu, Th 12:00PM - 1:15PM TBA Medford/Somerville	Fahad Rafique Dogar	3		
Meeting Dates		09/02/2025 - 12/08/2025					
Class Attributes		BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing					
Career		AS&E Undergrad/Grad					
Textbook		Textbook info					
Grading		Graded					
Enrollment Reqs		Completion of CS 15 and either CS 40 or EE 14, or Graduate Standing					
Repeat for Credit		No					
Allow Multiple Enroll in Term		No					
Total Units Allowed		3					
Total Completions Allowed		1					
Instruction Mode		In Person					
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		70	0	70			
Wait List		5	0	5			

CS-0115 **Database Systems**

Fundamental concepts of database systems, including conceptual design, relational and object-oriented data models, query languages (SQL, QBE), and implementation issues (indexing, transaction processing, concurrent control). The concepts and algorithms covered encompass many of those used in commercial and experimental database systems. Other topics include distributed databases and distributed query processing. Recommendations: CS 40

LECTURE							
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80263	Regular	Tu, Th 4:30PM - 5:45PM TBA Medford/Somerville	Johes Bater	3		
Meeting Dates		09/02/2025 - 12/08/2025					
Class Attributes		BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing					
Career		AS&E Undergrad/Grad					
Textbook		Textbook info					
Grading		Graded					
Enrollment Reqs		CS 15 or Graduate Standing					
Repeat for Credit		No					
Allow Multiple Enroll in Term		No					
Total Units Allowed		3					
Total Completions Allowed		1					
Instruction Mode		In Person					
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		70	0	70			
Wait List		5	0	5			
M1-LEC Details	82950	Regular	Mo 7:30PM - 9:00PM Online	Cody Doucette	3		☐

CS-0116 Introduction to Security

A holistic and broad perspective on cyber security. Attacking and defending networks, cryptography, vulnerabilities, reverse engineering, web security, static and dynamic analysis, malware, forensics. Principles illustrated through hands-on labs and projects, including Capture The Flag (CTF) games.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80264	Regular	Tu, Th 4:30PM - 5:45PM TBA Medford/Somerville	Ming Yan Chow	3		In Cart

Meeting Dates 09/02/2025 - 12/08/2025

Class Attributes BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing

Career AS&E Undergrad/Grad

Textbook [Textbook info](#)

Grading Graded

Enrollment Reqs

Requires the completion of CS 15 or graduate standing.

Repeat for Credit No

Allow Multiple Enroll in Term No

Total Units Allowed 3

Total Completions Allowed 1

Instruction Mode In Person

Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		100	2	98			
Wait List		10	0	10			

M1-LEC Details	82951	Regular	Th 5:30PM - 7:00PM Online	Ming Yan Chow	3		☐
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CS-0117 Internet-scale Distributed Systems

Principles and practices in designing large-scale distributed software systems on the Internet and beyond, including core principles of the design of the World-Wide Web. Key issues and fundamental principles are explored, e.g. global uniform naming, location independence, Metcalfe's law and network effects, function placement and the End-to-End principle, extensibility and evolution of distributed systems including Postel's law, leaky abstractions, etc. Comparison with more traditional distributed system designs, e.g. distributed objects, client/server, publish/subscribe, reliable queuing, and remote procedure calls.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80420	Regular	Tu, Th 4:30PM - 5:45PM TBA Medford/Somerville	Noah Mendelsohn	3		☒

Meeting Dates 09/02/2025 - 12/08/2025

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Class Attributes	SOE-Computing						
Career	AS&E Undergrad/Grad						
Textbook	Textbook info						
Grading	Graded						
Add Consent	Department Consent Required						

Enrollment Reqs
CS 40 or permission of the instructor

Class Notes
Please consult the high demand guide at <https://engineering.tufts.edu/cs/current-students/undergraduate/high-demand-enrollment> for information about registering for this high-demand course

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	3
Total Completions Allowed	1
Instruction Mode	In Person

Class Availability	Capacity	Enrolled	Available Seats
Enrollment	30	0	30
Wait List	0	0	0

CS-0118 **Cloud Computing**

Cloud computing fundamentals, including cloud architecture, scalability, elasticity, and metrics of cloud performance including service-level objectives (SLOs) and service-level agreements (SLAs). Cloud programming models and abstractions including Map/Reduce. Persistent storage mechanisms, including key/value stores and cold storage. Geo-distributed cloud systems. Cloud networking, including data center architecture, software defined networking, and middleboxes. Cloud security.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80421	Regular	Mo, We 4:30PM - 5:45PM TBA Medford/Somerville	Raja Raman Sambasivan	3		

Meeting Dates 09/02/2025 - 12/08/2025

Class Attributes	SOE-Computing
Career	AS&E Undergrad/Grad
Textbook	Textbook info
Grading	Graded

Enrollment Reqs
Completion of CS 40 or Graduate Standing

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	3
Total Completions Allowed	1
Instruction Mode	In Person

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		40	0	40			
Wait List		15	0	15			

CS-0119 **Big Data**

Collecting, processing, analyzing and acting on data at Internet scale with speed and complexity. Techniques and infrastructures developed for big data including parallel and distributed database systems, map-reduce infrastructures, scalable platforms for complex data types, stream processing systems, and cloud-based computing. Course blends theory, algorithms and practical (hands on) work.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC	82959	Regular	Mo 5:30PM - 7:00PM Online	Andrew Scott Winslow	4		
Details							

Meeting Dates	09/02/2025 - 12/18/2025
Class Attributes	SOE-Computing
Career	AS&E Graduate Online
Textbook	Textbook info
Grading	Graded

Enrollment Reqs
None

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	4
Total Completions Allowed	1
Instruction Mode	Online

Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		40	0	40			
Wait List		5	0	5			

CS-0120 **Web Programming and Engineering**

Web applications as complex systems that deliver functionality to a large number of users, and exhibit unique behaviors and demands in terms of performance, scalability, usability, and security. How the web works, limitations of client-side and server-side technologies including frameworks and APIs, content optimization, and data persistence and storage. Projects will involve search, using the cloud infrastructure, location-based services, mobile web development, and using tools to assess the security and privacy of web applications. Students may receive credit for either CS20 or CS120, but not both.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC	82956	Regular	Time Not Specified Online	Lisa Diorio	3		
Details							

Meeting Dates	09/02/2025 - 12/18/2025
Class Attributes	BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing
Career	AS&E Graduate Online
Textbook	Textbook info

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Grading	Graded						
Enrollment Reqs							
Open to online CS or DS postbac, OR online CS or DS MS students							
Repeat for Credit	No						
Allow Multiple Enroll in Term	No						
Total Units Allowed	3						
Total Completions Allowed	1						
Instruction Mode	Online						
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		30	0	30			
Wait List		5	0	5			

CS-0121 **Software Engineering**

Core principles and ideas that enable development of large-scale software systems, with a focus on programming. Abstraction, modularity, design patterns, specification, testing, verification, and debugging.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80422	Regular	Mo, We 3:00PM - 4:15PM TBA Medford/Somerville	Jeffrey Foster	3		In Cart
Meeting Dates 09/02/2025 - 12/08/2025							
Class Attributes BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing							
Career AS&E Undergrad/Grad							
Textbook Textbook info							
Grading Graded							
Enrollment Reqs Requires the completion of CS 15 or graduate standing.							
Repeat for Credit No							
Allow Multiple Enroll in Term No							
Total Units Allowed 3							
Total Completions Allowed 1							
Instruction Mode In Person							
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		150	0	150			
Wait List		0	0	0			
M1-LEC Details	82952	Regular	We 6:00PM - 7:30PM Online	Jeffrey Foster	3		

CS-0131 **Artificial Intelligence**

History, theory, and computational methods of artificial intelligence. Basic concepts include representation of knowledge and computational methods for reasoning. One or two application areas will be studied, to be selected from

expert systems, robotics, computer vision, natural language understanding, and planning.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80266	Regular	Mo, We 6:00PM - 7:15PM TBA Medford/Somerville	STAFF	3		

Meeting Dates	09/02/2025 - 12/08/2025
Class Attributes	BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing
Career	AS&E Undergrad/Grad
Textbook	Textbook info
Grading	Graded

Enrollment Reqs
CS 15 and (CS/Math 61 or Math 65) or Grad Standing

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	3
Total Completions Allowed	1
Instruction Mode	In Person

Class Availability	Capacity	Enrolled	Available Seats
Enrollment	60	0	60
Wait List	10	0	10

M1-LEC Details	82957	Regular	Mo 7:30PM - 9:00PM Online	Fabrizio Santini	3		
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CS-0132 **Computer Vision**

Introduction to low and intermediate levels of classic and modern Computer Vision. How to design algorithms that process visual scenes to automatically extract information. Fundamental principles and important applications of computer vision, including image formation, processing, detection and matching features, image segmentation, and multiple views. Basics of machine learning and deep learning for computer vision. Recommendations: CS 160 and Math 165

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80452	Regular	Mo, We 6:00PM - 7:15PM TBA Medford/Somerville	Roy Shilkrot	3		

Meeting Dates	09/02/2025 - 12/08/2025
Class Attributes	SOE-Computing
Career	AS&E Undergrad/Grad
Textbook	Textbook info
Grading	Graded

Enrollment Reqs
Completion of CS 15, Math 32, Math 70 and one of (Math 166 or ES 56 or EE 24) or graduate standing

Repeat for Credit	No
Allow Multiple Enroll in Term	No

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Total Units Allowed	3						
Total Completions Allowed	1						
Instruction Mode	In Person						
Class Availability	Capacity	Enrolled	Available Seats				
Enrollment	40	0	40				
Wait List	5	0	5				

CS-0133 **Human-Robot Interaction**

Human-robot interaction (HRI) at the intersection of psychology, human factors engineering, computer science, and robotics. Case studies of developing robots for various societal tasks, from flexible and adaptive robots for industrial manufacturing, to socially assistive robots for elder care. Discussion of key papers in HRI. Short student presentations on HRI studies and designs; work in interdisciplinary groups on a term project to design and conduct an HRI study. Recommendations: At least one course beyond Math 34, including Math 42, 44, 61, 70, or 72

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80423	Regular	Mo, We 3:00PM - 4:15PM TBA Medford/Somerville	Elaine Schaertl Short	3		

Meeting Dates	09/02/2025 - 12/08/2025
Class Attributes	SOE-Computing
Career	AS&E Undergrad/Grad
Textbook	Textbook info
Grading	Graded

Enrollment Reqs
Prior completion of CS 15 andn senior standing, or graduate standing

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	3
Total Completions Allowed	1
Instruction Mode	In Person

Class Availability	Capacity	Enrolled	Available Seats
Enrollment	45	0	45
Wait List	5	0	5

CS-0134 **Computational Models in Cognitive Science**

(Cross-listed w/ PSY 141) Scientific logic of using computational models for testing theories in cognitive science. Connectionist and Bayesian models; agent-based simulation. Emphasis upon using models in combination with empirical data to test theories. Appropriate use and critical evaluation of computational modeling as found in scientific publications. Recommendations:CS 10, 11, or equivalent programming experience in any computer language. An introductory course in statistics.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80424	Regular	Tu, Th 9:00AM - 10:15AM TBA	Johannes Peter Albert De Ruiter	3		
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Medford/Somerville

Meeting Dates	09/02/2025 - 12/08/2025
Class Attributes	SOE-Computing
Career	AS&E Undergrad/Grad
Textbook	Textbook info
Grading	Graded

Enrollment Reqs
None

Class Notes
Proficiency in at least one programming language required

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	3
Total Completions Allowed	1
Instruction Mode	In Person

Class Availability	Capacity	Enrolled	Available Seats	
Enrollment	32	0	32	
Wait List	5	0	5	

Combined Section	Description	Enr Total	Wait Total	Status
CS-0134-01 LEC (80424)	Computational Models Cog Sci	0	0	
PSY-0141-01 LEC (83049)	Computational Models Cog Sci	0	0	

CS-0135 Introduction To Machine Learning And Data Mining

An overview of methods whereby computers can learn from data or experience and make decisions accordingly. Topics include supervised learning, unsupervised learning, reinforcement learning, and knowledge extraction from large databases with applications to science, engineering, and medicine. Recommendations: CS 160 is highly recommended.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80268	Regular	Tu, Th 9:00AM - 10:15AM TBA Medford/Somerville	STAFF	3		☐

Meeting Dates	09/02/2025 - 12/08/2025
Class Attributes	BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing
Career	AS&E Undergrad/Grad
Textbook	Textbook info
Grading	Graded

Enrollment Reqs
CS 15 and (CS/Math 61 or Math 65) or Grad Standing

Repeat for Credit	No
Allow Multiple Enroll in Term	No

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Total Units Allowed	3						
Total Completions Allowed	1						
Instruction Mode In Person							
Class Availability	Capacity	Enrolled	Available Seats				
Enrollment	105	1	104				
Wait List	15	0	15				
M1-LEC	82953	Regular	Th 5:30PM - 7:00PM Online	Chris Magnano	3		☐
Details							

CS-0136 **Statistical Pattern Recognition**

Statistical foundations and algorithms for machine learning with a focus on Bayesian modeling. Topics include: classification and regression problems, regularization, model selection, kernel methods, support vector machines, Gaussian processes, graphical models. Recommendations: MATH 70 or 72, EE 104 or MATH 165, and CS 40 or CS 105, or graduate standing

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	83160	Regular	Tu, Th 1:30PM - 2:45PM TBA Medford/Somerville	Michael C. Hughes	3		☒
Details							

Meeting Dates 09/02/2025 - 12/08/2025

Class Attributes SOE-Computing

Career AS&E Undergrad/Grad

Textbook [Textbook info](#)

Grading Graded

Enrollment Reqs

Graduate standing OR Major in (Data Science OR Computer Science) in SOE OR Major in (Computer Science) in A&S

Repeat for Credit No

Allow Multiple Enroll in Term No

Total Units Allowed 3

Total Completions Allowed 1

Instruction Mode In Person

Class Availability	Capacity	Enrolled	Available Seats
Enrollment	40	0	40
Wait List	10	0	10

CS-0138 **Reinforcement Learning**

Agents that learn, plan, and act in complex, uncertain environments. Varying topics that may include practical successes and challenges of reinforcement learning; common software libraries and packages used to implement and test reinforcement learning algorithms, and advanced topics such as transfer learning and deep reinforcement learning. Recommendations: CS 131 and/or CS 135. Proficiency in one of C++, Java, or Python.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select												
01-LEC Details	83163	Regular	Tu, Th 10:30AM - 11:45AM TBA Medford/Somerville	Jivko Sinapov	3														
Meeting Dates 09/02/2025 - 12/08/2025 Class Attributes SOE-Computing Career AS&E Undergrad/Grad Textbook Textbook info Grading Graded																			
Enrollment Reqs CS 15 and (CS/Math 61 or Math 65) or Grad Standing																			
Class Notes Recommendations: CS 15 and either CS 61/MATH 61 or MATH 65; or graduate standing; or permission of instructor																			
Repeat for Credit No Allow Multiple Enroll in Term No Total Units Allowed 3 Total Completions Allowed 1 Instruction Mode In Person																			
<table><tr><th>Class Availability</th><th>Capacity</th><th>Enrolled</th><th>Available Seats</th></tr><tr><td>Enrollment</td><td>65</td><td>0</td><td>65</td></tr><tr><td>Wait List</td><td>10</td><td>0</td><td>10</td></tr></table>								Class Availability	Capacity	Enrolled	Available Seats	Enrollment	65	0	65	Wait List	10	0	10
Class Availability	Capacity	Enrolled	Available Seats																
Enrollment	65	0	65																
Wait List	10	0	10																
M1-LEC Details	82961	Regular	We 7:00PM - 8:30PM Online	STAFF	3														
CS-0150	Special Topics - Goal and Plan Recognition																		
CS-0150	Special Topics - Advanced Programming Languages																		
CS-0150	Special Topics - Quantum Computer Science																		
CS-0150	Special Topics - Generative Models																		
CS-0150	Special Topics - AI Foundations																		
CS-0150	Special Topics - AI Planning																		
CS-0150	Special Topics - Algorithmic Music Composition																		
CS-0150	Special Topics - Scientific Machine Learning																		
CS-0151	Special Topics in Data Infrastructure and Systems - Software Perf Eng																		
CS-0160	Algorithms																		
Introduction to the study of algorithms. Strategies such as divide-and-conquer, greedy methods, and dynamic programming. Graph algorithms, sorting, searching, integer arithmetic, hashing, and NP-complete problems. High demand (see "course notes" for signup procedure).																			
LECTURE																			
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select												
M1-LEC Details	82954	Regular	Th 7:00PM - 9:00PM Online	STAFF	4														
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select												

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Meeting Dates	09/02/2025 - 12/18/2025						
Class Attributes	BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing						
Career	AS&E Graduate Online						
Textbook	Textbook info						
Grading	Graded						

Enrollment Reqs
Open to online CS or DS postbac, OR online CS or DS MS students

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	4
Total Completions Allowed	1
Instruction Mode	Online

Class Availability	Capacity	Enrolled	Available Seats
Enrollment	40	0	40
Wait List	0	0	0

CS-0160 **Algorithms**

Introduction to the study of algorithms. Strategies such as divide-and-conquer, greedy methods, and dynamic programming. Graph algorithms, sorting, searching, integer arithmetic, hashing, and NP-complete problems. High demand (see "course notes" for sign up procedure).

LECTURE					Select 1 Lecture, 1 Recitation		
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	80270	Regular	Mo, We 4:30PM - 5:45PM TBA Medford/Somerville	Karen Edwards	4		☐
Details							

Meeting Dates	09/02/2025 - 12/08/2025						
Class Attributes	BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing						
Career	AS&E Undergrad/Grad						
Textbook	Textbook info						
Grading	Graded						

Enrollment Reqs
Requires prior completion of CS 15 with a grade of C- or better and a one of (CS 61, Math 61, or Math 65) with a C- or better, or graduate standing. First years and sophomores must have completed CS 15 and Math 61/CS 61/Math 65 for a B+ or better.
Computer Science or Data Science Majors or Postbac or Grad Students

Class Notes
Students who do not meet the enrollment requirements can request permission by signing up for the special waitlist section (lecture section 02 and lab/recitation section LW/RW). Priority will be given to CS minors. Please consult the course registration guide at <https://engineering.tufts.edu/cs/current-students/undergraduate/high-demand-enrollment> for information about registering for this high-demand course.

Repeat for Credit	No
Allow Multiple Enroll in Term	No
Total Units Allowed	4

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Total Completions Allowed	1						
Instruction Mode	In Person						
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		160	1	159			
Wait List		20	0	20			

RECITATION

CS-0163 Computational Geometry

LECTURE

CS-0165 Cryptography

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Medford/Somerville							
CS-0166	Computational Systems Biology						
(Cross-listed as BME 144) Computational modeling of complex biological systems to analyze their emergent properties and understand or predict system behavior. Molecular and protein modeling in the context of biochemical networks. Application of these models across a select set of applications such as metabolomics, gut microbiome analysis, modularity analysis, flux balance analysis, and biological discovery. Introductions to machine learning, linear systems, and statistical concepts that are commonly used in systems biology.							
LECTURE							
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	80479	Regular	Mo, We 10:30AM - 11:45AM TBA Medford/Somerville	Soha Hassoun	3		
Details							
Meeting Dates		09/02/2025 - 12/08/2025					
Class Attributes		SOE-Computing					
Career		AS&E Undergrad/Grad					
Textbook		Textbook info					
Grading		Graded					
Enrollment Reqs							
CS 15 and at least one CS class numbered above 100 or graduate standing							
Repeat for Credit		No					
Allow Multiple Enroll in Term		No					
Total Units Allowed		3					
Total Completions Allowed		1					
Instruction Mode		In Person					
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		30	0	30			
Wait List		5	0	5			

CS-0170 **Computation Theory**

(Cross-listed as MATH 191). Models of computation: Turing machines, pushdown automata, and finite automata. Grammars and formal languages, including context-free languages and regular sets. Important problems, including the halting problem and language equivalence theorems.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
M1-LEC Details	82955	Regular	Tu 7:30PM - 9:00PM Online	Andrew Scott Winslow	3		

Meeting Dates

09/02/2025 - 12/18/2025

Class Attributes

BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing

Career

AS&E Graduate Online

Textbook

[Textbook info](#)

Grading

Graded

Enrollment Reqs
Open to online CS or DS postbac, OR online CS or DS MS students

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
---------	-----------	---------	--------------------------	---------	--------	--------	--------

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Repeat for Credit	No						
Allow Multiple Enroll in Term	No						
Total Units Allowed	3						
Total Completions Allowed	1						
Instruction Mode	Online						
Class Availability	Capacity	Enrolled	Available Seats				
Enrollment	40	0	40				
Wait List	5	0	5				

CS-0170 Computation Theory

(Cross-listed as MATH 191). Models of computation: Turing machines, pushdown automata, and finite automata. Grammars and formal languages, including context-free languages and regular sets. Important problems, including the halting problem and language equivalence theorems.

LECTURE

Select 1 Lecture, 1 Recitation

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	80271	Regular	Tu, Th 1:30PM - 2:45PM TBA Medford/Somerville	Lenore J Cowen	3		☐
Details							

RECITATION

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
RA-RCT	80025	Regular	We 9:00AM - 10:15AM TBA Medford/Somerville	Lenore J Cowen	0		☐
Details							
RB-RCT	80026	Regular	We 12:00PM - 1:15PM TBA Medford/Somerville	Lenore J Cowen	0		☐
Details							
RC-RCT	80208	Regular	We 3:00PM - 4:15PM TBA Medford/Somerville	Lenore J Cowen	0		☐
Details							
RD-RCT	80027	Regular	We 4:30PM - 5:45PM TBA Medford/Somerville	Lenore J Cowen	0		☐
Details							
RE-RCT	80028	Regular	We 6:00PM - 7:15PM TBA Medford/Somerville	Lenore J Cowen	0		☐
Details							

CS-0175 Computer Graphics

The course provides the background to understand and build interactive graphics systems. Introduction to computer graphics from simple two-dimensional graphics through three-dimensional viewing and transforms. Interactive graphics programming, raster graphics, raster algorithms, geometrical transformations, viewing in three dimensions, modeling and hierarchical data structures, visual realism, virtual reality, interaction devices, tasks and techniques. Final project. Recommendations: MATH 42 or 44, and MATH 70 or 72, or graduate standing.

LECTURE

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC	80426	Regular	Tu, Th 12:00PM - 1:15PM TBA Medford/Somerville	Remco K Chang	3		☒
Details							

Meeting Dates 09/02/2025 - 12/08/2025

Class Attributes BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing

Career AS&E Undergrad/Grad

Textbook [Textbook info](#)

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
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Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Grading	Graded						
Enrollment Reqs							
Completion of CS 40 or CS/DS/SSD Graduate Standing							
Repeat for Credit							
No							
Allow Multiple Enroll in Term							
No							
Total Units Allowed							
3							
Total Completions Allowed							
1							
Instruction Mode							
In Person							
Class Availability							
		Capacity	Enrolled	Available Seats			
Enrollment		40	0	40			
Wait List		5	0	5			

CS-0191	Research
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CS-0193	Directed Study
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Guided study of an approved topic. Please see departmental website for specific details.

INDEPENDENT STUDY							
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-IND Details	80273	Regular	Time Not Specified TBA Medford/Somerville	Abani Patra, Alva Couch, Chris Magnano, Daniel Jared Votipka, Dave Lillethun, Deborah Sunter, Diane L Souvaine, Donna Slonim, Elaine Schaertl Short, Elyse L. Cornwall, Fahad Rafique Dogar, Guannan Wei, Hari Sundar, Jeffrey Foster, Jivko Sinapov, Johannes Peter Albert De Ruiter, Johes Bater, Karen A Panetta, Karen Edwards, Lenore J Cowen, Liping Liu, Mark A Sheldon, Marty Allen, Matthias Scheutz, Megan Monroe, Megumi Ando, Michael C. Hughes, Milod Kazerounian, Ming Yan Chow, Noah Mendelsohn, Peter John Love, Raja Raman Sambasivan, Remco K Chang, Reuth Mirsky, Richard Townsend, Robert Jacob, Saeed Mehraban, Soha Hassoun, Susan Landau, Vladimir Podolskii	0-4		

Meeting Dates		09/02/2025 - 12/08/2025					
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Class Attributes							
BFA-Science/Technology, LA-Distribution-Mathematics, SOE-Computing							
Career	AS&E Undergrad/Grad						
Textbook	Textbook info						
Grading	Graded						
Add Consent	Department Consent Required						
Enrollment Reqs							
None							
Repeat for Credit	Yes						
Allow Multiple Enroll in Term	Yes						
Total Units Allowed	99						
Total Completions Allowed	99						
Instruction Mode	Advisor Consultation						
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		50	0	50			
Wait List		0	0	0			

CS-0195 **Cybersecurity Clinic I**

Students work in teams on year-long open-ended projects for real clients. Password management, asset tracking, incident response planning, business continuity planning, risk mitigation, source code review, vulnerability assessments, penetration testing, and training. Cybersecurity Clinic is a member of the The Consortium of Cybersecurity Clinics. Students who take CS 195 will continue working on the same project in CS 196 the following semester. Additional requirements apply. See Course Notes for details on applying to the Cybersecurity Clinic program and signing up for the course.

INTERNSHIP

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-INT Details	84422	Regular	Time Not Specified TBA Medford/Somerville	Ming Yan Chow	3		
Meeting Dates 09/02/2025 - 12/08/2025							
Class Attributes SOE-Computing							
Career AS&E Undergrad/Grad							
Textbook Textbook info							
Grading Graded							
Add Consent Department Consent Required							
Enrollment Reqs A- in CS 116, CS 114, CS 28, or PS 188 and Junior or Senior Standing, or Grad							
Repeat for Credit No							
Allow Multiple Enroll in Term No							
Total Units Allowed 3							
Total Completions Allowed 1							
Instruction Mode In Person							
Class Availability Capacity Enrolled Available Seats Enrollment 20 0 20							
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Class Availability		Capacity	Enrolled	Available Seats			
CS-0197	Honors Thesis-computer Science A						
CS-0201	Cyber for Future Policymakers						
(Cross-listed w/ DHP D291) Graduate version of CS 12. Relevance of computer technologies to policy development. Internet architecture and basic networking, the Web, cloud architectures, cryptography, security and privacy, artificial intelligence and machine learning, and open-source systems. Developing technologies, including quantum computing and post-quantum cryptography. Prerequisites: Graduate standing in a discipline other than Computer Science, Data Science, Bioinformatics, Cognitive Science, or Human-Robot Interaction. Recommendation: An introductory course in computer science including programming in any language, roughly equivalent to CS 10 or 11.							
LECTURE							
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80231	Regular	Tu, Th, Fr 10:30AM - 11:45AM TBA Medford/Somerville	Dave Lillethun	3		
CS-0202	How Systems Work						
(Cross-listed w/ DHP D292) Graduate version of CS 13. How computing systems work: bits, bytes, the representation of information, the CPU, assembly language, programming languages. Networking: including peering, packets, and the Internet. Algorithms and the fundamental limitations of computing. Recommendations: An introductory course in computer science including programming in any language, roughly equivalent to CS 10 or 11.							
LECTURE							
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
01-LEC Details	80232	Regular	Tu, Th, Fr 10:30AM - 11:45AM TBA Medford/Somerville	Mark A Sheldon	3		
Meeting Dates	09/02/2025 - 12/08/2025						
Career	AS&E Graduate						
Textbook	Textbook info						
Grading	Graded						
Enrollment Reqs		Open only to CSPP and Fletcher					
Class Notes		Completion of CS 10 or CS 11 recommended					
Repeat for Credit	No						
Allow Multiple Enroll in Term	No						
Total Units Allowed	3						
Total Completions Allowed	1						
Instruction Mode	In Person						
Class Availability		Capacity	Enrolled	Available Seats			
Enrollment		35	0	35			
Wait List		5	0	5			
Combined Section		Description	Enr Total	Wait Total	Status		
CS-0202-01 LEC (80232)		How Systems Work	0	0			
Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select

Section	Class No.	Session	Day, Times and Locations	Faculty	Credit	Status	Select
Combined Section	Description		Enr Total	Wait Total	Status		
CS-0288	Master of Science Capstone Project I						
CS-0289	Master of Science Capstone Project II						
CS-0291	Seminar in Computer Science						
CS-0293	Master's Project						
CS-0295	Masters Thesis						
CS-0297	Dissertation Research						
CS-0299	Internship In Computer Science						
CS-0401	Masters Degree Continuation						
CS-0402	Masters Degree Continuation						
CS-0404	Graduate Engineering Co-op						
CS-0405	Grad Teaching Assistant						
CS-0406	Grad Research Assistant						
CS-0501	Doctoral Degree Continuation						
CS-0502	Doctoral Degree Continuation						

Academic Policies

The Academic Policies section of this handbook includes general principles and practices of the School of Engineering. The following are the minimum requirements to maintain good academic standing. Departments may have more stringent requirements. To be informed of specific academic policies of your department or program, consult your departmental or graduate program guidelines.

Academic Standing

All graduate students are expected to remain in good academic standing. In order to remain eligible to continue in a degree, certificate, or post-baccalaureate program and/or to receive various types of financial assistance, including federal aid, a graduate student must maintain good academic standing and be making satisfactory progress toward the degree or certificate. The faculty has adopted the following statements relative to academic standing and progress toward a certificate or degree.

Note: Honors standing is not given in the graduate programs. The following are the minimum requirements; departments have the right to impose additional criteria for good academic standing.

To remain in good academic standing a student may have only one grade lower than B-, or one Unsatisfactory (U), or one Incomplete (I) that remains incomplete for more than one semester, or Permanent Incomplete (PI).

1. Credit will be awarded only to credit-granting courses that receive a grade of B- or better or satisfactory (S).
2. Options for making up a grade less than B- or a U must be discussed and approved by the department. Courses for which a student receives a grade lower than B- may be retaken only once to achieve the credit. The original grade earned remains in the student's record.
3. Incomplete grades must be completed in accordance with the Policy on Incompletes detailed in this handbook on page 6.
4. Funded MS/Ph.D. or Ph.D. students must receive satisfactory performance evaluations on their graduate, research or teaching assistant duties, each semester or can be subject to immediate dismissal.
5. Deviation from any of the above requirements or the departmental or program requirements constitutes evidence that the student is making inadequate progress toward a certificate or degree and is no longer in good academic standing.

Dismissal

The student will be officially notified of their status and thereupon the student will be dismissed unless recommended otherwise by the department and an exception is granted by the graduate dean. The student will be informed of the recommendation. Only the Dean of Graduate Education for the School of Engineering may dismiss an enrolled graduate student. In general, dismissal occurs following the review and vote of the department's graduate program committee. Students will be kept informed throughout these proceedings. The dismissal takes effect immediately upon receiving notification.

All students may appeal the decision in writing, within five days of receiving notification of dismissal, to the dean of graduate education. The appeal needs to be submitted within five business days from the date of the withdrawal notification. Submit the appeal in writing via email to the Dean of Graduate Education, karen@ece.tufts.edu, copy to enggradstudies@tufts.edu, academic advisor, the department chair, and the program director. Specify the reason for reconsideration. If an appeal or an exception is granted, the graduate director or the department chair will meet with the student to develop a plan and a timeline to return to good academic standing. The set plan and timeline must be communicated to the dean's office.

Students enrolled in the Fifth-Year MS program must consistently maintain good academic standing, or risk having their admission into the master's program rescinded.

Fifth-Year MS program students must complete and earn their undergraduate degree before matriculating into the MS program. The degrees cannot be completed concurrently.

Among the most common grounds for dismissal from the institution are the following:

- More than one grade not meeting the minimum standards described
- Failure(s) to reach a departmental benchmark
- Evidence that degree requirements will not be met within stated time limits
- Academic dishonesty (more information on the Academic Integrity Policy is below)
- Violation of the code of conduct or working together policy

Academic Integrity

Students are expected to meet the highest standards of academic integrity. Plagiarism, fabrication, falsification, cheating, and other acts of academic dishonesty, or abetting the academic dishonesty of another will result in sanctions. The Tufts [Academic Integrity Policy](#) provides a primer on ethical academic work. It also includes a description of computer ethics and the appropriate use of electronic resources. Students should familiarize themselves with the standards outlined in this policy.

Alleged violations of these policies will be considered and addressed within the [Student Conduct Resolution Procedure](#).

Academic Grievances

Graduate students who are experiencing conflicts in their academic work may approach for consultation the office of the Dean of Graduate Education for the School of Engineering. The dean will consult with students on how to address academic conflicts with faculty or with other graduate students in their departments. Various options for conflict management, including mediation, will be explored.

Policy on Incompletes

An incomplete grade may be awarded only if the student has done work in the course, the instructor judges the reasons for granting incomplete status to be valid, and the instructor determines that the work can be completed in the time specified on the Incomplete Form, available in SIS. The Incomplete form may be found on the faculty SIS homepage under “Faculty Forms”. It is the responsibility of the student to request an incomplete before the due date of the required work and sign the contract with the instructor of the course prior to the grading deadline for the term the incomplete is to be assigned. The instructor is responsible for specifying on the form the deadline for submission of work and the conditions that the student must satisfy to receive a grade.

If an incomplete is granted, all work in the course must be completed on or before the date six weeks after the first day of classes in the subsequent semester (fall or spring only; summer terms excluded). Instructors who wish to set a later deadline may do so, subject to approval from the Director of Graduate Study in the program where the student is enrolled. If the student has completed the work within the stated time, it will be evaluated without prejudice.

Residency Requirements

For doctoral programs and the master’s programs in Cybersecurity and Public Policy, the residency requirement is two semesters, excluding summer of graduate study at Tufts. All SOE MS degree program students must be in residence (enrolled as Tufts SOE graduate students) for a minimum of 60% of their required graduate program credits, exclusive of graduate seminars for the specific program they are enrolled in. This rule also applies to dual-degree and fifth year students.

Tufts will not award the doctoral degree to students whose dissertation research or writing was performed at another institution, unless they were under the direct supervision of a Tufts-based faculty member.

Registration

Continuous Enrollment Policy

Graduate students must be enrolled (registered), or on an approved leave of absence, for every academic-year semester between matriculation and graduation. Graduate students may only register for courses that count toward their degree program. Students should register during the early registration periods in November and April. The university reserves the right to withhold registration for any student having unmet Academic Integrity Training (AIT), financial, or health services obligations. Students who fail to register by the end of the first week of classes, or the add deadline for the term, whichever is first, will be administratively withdrawn and subject to a \$350 reinstatement fee. International students must maintain proper enrollment status per their visa requirements.

Degree-only Status

If a student has registered for all required courses, including thesis research (295, 296) or dissertation research (297, 298), the student must register for a course designation that indicates that only thesis, project, master's exhibition, or dissertation-related work, whether part-time or full-time, is being pursued.

Master's candidates must register for course 401-PT (part-time) or 402-FT (full-time) in their department and doctoral candidates must register for course 501-PT (part-time) or 502-FT (full-time) in their department. Graduate students may declare full-time status of thirty-five hours per week (402 or 502) with their advisor's concurrence.

Enrollment Status

Graduate students are responsible for maintaining enrollment status at Tufts.* Federal regulations require students to be enrolled (registered) full-time or half-time in order to receive and/or defer student loans. Tufts provides information regarding student enrollment status to lenders via the National Student Loan Clearinghouse and is required by law to return funds for students who do not maintain a minimum of half-time enrollment status. Enrollment status is either full-time, half-time, or part-time as defined below:

Full-time: Nine (9) or more course credits, six (6) course credits and a teaching or research assistantship (must also register for 405-TA or 406-RA), or registration as a full-time continuing student (402/502) working on a thesis, dissertation, project, or internship. Students admitted full-time who register for credits totaling less than full-time will not achieve a full-time status.

Half-time: Five (5) course credits; Two (2) course credits plus a teaching or research assistantship.

Part-time: One (1) to Four (4) course credits, or registration as a part-time continuing student (401/501).

**In most situations, international students must maintain full-time status. In certain circumstances, international students may be authorized for a [reduced course load](#), allowing them to drop below full-time: [Contact the International Center](#) with questions or concerns about Enrollment Status.*

Cross-Registration and Graduate School Consortium

During the academic year, full-time graduate students may take one course per semester through cross-registration agreements with Boston College, Boston University, and Brandeis University. A full-time graduate student at Tufts University may also enroll for two graduate courses during any semester at the Fletcher School of Law and Diplomacy, the Friedman School of Nutrition Science and Policy, and the Graduate School of Biomedical Sciences. Cross-registration is possible on a space available basis. Students who wish to cross-register should consult with the instructor of the course, and should expect to satisfy any prerequisites typically required for enrollment. Cross-registration is not permitted in any summer school. Courses satisfactorily completed (B– or better) at one of the three consortium schools (Boston College, Boston University, and Brandeis University) automatically appear on the student's Tufts transcript and may be counted toward degree requirements.

Tufts students who wish to cross register at a consortium school should first consult with their academic advisor and/or the department's graduate studies committee representative before completing the online [Cross-Registration Petition Form](#) through SIS. The host institution reserves the right to terminate the student's participation at that institution at any time.

Cross-registration is also offered through the Consortium for Graduate Studies in Gender, Culture, Women, and Sexuality at MIT. This consortium relationship is limited to the specific area of Women, Gender, and Sexuality Studies. For information about course offerings and application materials, visit the [GCWS website](#).

Audits

Graduate students may arrange with an instructor to sit in on a course, but this course will not appear on the academic record.

Dropping a Course

A course for which a student has registered remains on the record unless it has been dropped within the first five weeks of the term. Courses that are dropped after the fifth week but prior to the last day of classes will carry the grade of W and remain on the transcript. Students who are billed per credit should review the refund policy on the bursar's website for details: <https://students.tufts.edu/financial-services/billing/tuition-refund-policy/>.

Grades

Grades of scholarship are expressed by one of the following letters:

- A Superior work
- B Meritorious work
- C Not acceptable for graduate credit
- D Not acceptable for graduate credit
- F Failure
- P Not acceptable for graduate credit
- S, U Grades of S (Satisfactory) and U (Unsatisfactory) may be given by the instructor in special topics courses, courses in supervised teaching, research courses, certain graduate colloquia, certain professional development courses such as Graduate Institute for Teaching (GIFT), Graduate Research Excellence at Tufts (GREAT), thesis courses, and dissertation courses.

The following symbols are also used:

- I Incomplete: an indication that more time will be allowed to complete the work, specifically within six weeks of the first day of classes in the subsequent semester (fall or spring only; summer terms excluded).
- PI Permanent Incomplete: Students who received an Incomplete and do not complete the work within the stated time will receive a Permanent Incomplete (PI).
- W Withdrawn: an indication that a student has been permitted to withdraw from a course after the fifth week of a semester, but no later than the last day of classes.
- Y Work not scheduled for evaluation during the current term.

Changes in Course Grades: Statute of Limitations

Effective education requires timely and objective evaluation of students' academic work, using clear, standard, fair, and public criteria. Such standards should be listed

Graduate Student Handbook Supplement

*If you need further clarification, please contact the Director of Graduate Studies
Prof. Diane Souvaine at diane.souvaine@tufts.edu*

*or the Graduate Program Manager
Johnny Redmond at johnny@cs.tufts.edu*

*or the Graduate Program Coordinator
Sandie Schulenburg at Sandra.Schulenburg@tufts.edu*

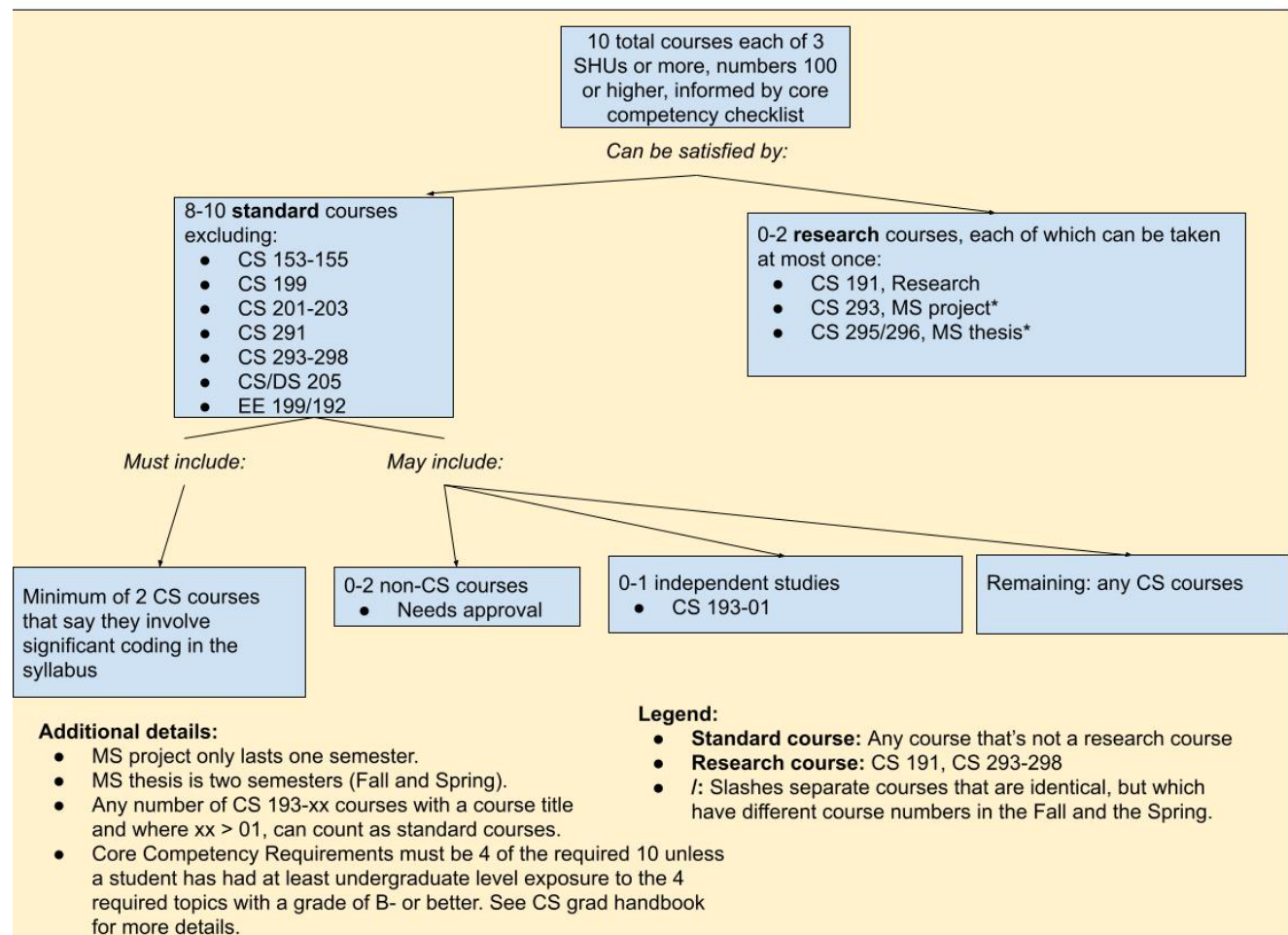
Updated 12/02/2024

Tufts course requirements for a Master's and Ph.D. in Computer Science

This document lists important information about course requirements for Master's and Ph.D. Students. The term “courses” refers to lecture-based classes, independent studies, and research. The first two are denoted by “standard courses” and the latter by “research courses. Both Master's and Ph.D. course requirements can be satisfied via a varying combination of standard and research courses depending on your interests.

We recommend that Master's students interested in completing a thesis and Ph.D. students bias their course selection toward research courses.

M.S. in Computer Science (10 total courses, each of 3 SHUs or more)



The flow chart above illustrates the course requirements to get a master's degree. A box that indicates a range in required courses (e.g., 8-10 standard courses) indicates that some of the required courses can be obtained from a box in a sibling branch (e.g., 0-2 research courses).

Core Competencies: By your last semester at Tufts, you must have completed at least one class in each of the four areas listed in Appendix E of the handbook and reproduced below. Designated faculty will hold core competency certification sessions during the first seven days of each semester and can approve and/or advise you on the completion of this requirement.

The competencies can be filled by equivalent classes you may have taken at other universities, and that appear on that university's transcript. Alternatively, you can fill them at Tufts by the courses listed in the sub-bullets below. You must have earned at least a B- in a course, whether at Tufts or elsewhere, to satisfy the relevant course-competency requirement. You will not receive graduate course credit for any course numbered less than 100.

Core Competency areas include:

- Computer Architecture and Assembly Language (CA&AL)
 - CS 40, Machine Structure. *No graduate credit.*
 - CS 111, Operating Systems
 - CS 112, Networks
 - CS 114, Network Security
 - CS 116, Introduction to Security
 - CS 118, Cloud Computing
 - CS 146, (also EE 126) Computer Engineering
 - CS 107 (Formerly COMP/CS 181), Compilers; offered infrequently
 - CS 140, Advanced Topics in Computer Architecture
- Programming Languages (PL)
 - CS 105, Programming Languages
 - CS 21, Concurrent Programming. *No graduate credit.*
 - CS 86, Object-Oriented Programming for GUIs. *No graduate credit.*
 - CS 121 (Formerly COMP/CS 180), Software Engineering
 - CS 107 (Formerly COMP/CS 181), Compilers; offered infrequently
- Data structures and Analysis of Algorithms (DS&AA)
 - CS 160: Intro to Algorithms (we highly recommend taking this class!)
- Theory of Computation (ToC)
 - CS 170, Computation theory
 - If you have little math background, try to take Discrete Math (COMP/CS 61) first.

CS 191: This course is a vehicle for doing research. It has similar requirements to the M.S. project (see below). This course can be taken at most once.

CS 199 (Internship in Computer Science): This course is a vehicle for international students to complete an internship. It does not count towards the 10 course requirement. Reach out to Professor Ming Chow for more information on this course.

M.S. Thesis: The thesis requires a commitment of two semesters total, recorded by enrolling in CS 295 and CS 296 in either order; the M.S. thesis is completely optional. Acceptance to the thesis track occurs after matriculation into the program and only with the support of a faculty advisor who is interested in supervising thesis work. After finding a faculty member who is willing to work with you on a MS Thesis, the faculty member can write to the CS Graduate coordinator to request your change to the thesis track. Some reasons for deciding to do a M.S. thesis may include: 1) you are a M.S. student who wants significant research experience; 2) You want to leave the Ph.D. program with a master's and retain some official record of your research activity; 3) You want to complete a substantial and polished preliminary research project on the way to a Ph.D. You need to submit a thesis prospectus at the end of the first semester and the thesis document just after the end of classes during the semester in which you defend. The deadline can be confirmed [here](#). The defense should be scheduled two weeks earlier than the deadline.

The student and advisor will jointly select a thesis committee subject to approval by the CS Graduate Committee. This committee must include at least three faculty members, including one member from outside the department. You will defend the research via a 45-minute presentation, which the committee will attend.

The defense is also open to the public. You will receive a grade for CS 295 and CS 296 only after you finish both semesters.

M.S. Project: An M.S. Project consists of research conducted with a faculty advisor and is a commitment of one semester recorded by registering for CS 293, usually for 3 SHUs. You may choose to take this option because: 1) you are a M.S. student who wants to complete a research project without the overhead of writing a thesis, or 2) you are a Ph.D. student who doesn't want to write a separate master's thesis. If you do not have an advisor for the project, you need to find one. If your advisor for the project is not your academic advisor, they need to agree to become your advisor. You and your advisor need to agree on the project and any write-up requirements. (Advisors approve the write-up and keep a copy). This course can be taken at most once.

CS 193-xx (where $xx > 01$): These are directed study courses with official names and with two or more students. Your advisor might create one as a vehicle for reading a set of research papers from a conference or understanding a new research area. Another faculty member might create one that involves a joint implementation project, or as a dry run of a course that will become a CS 150. You can count any number of these as *standard* CS courses.

Maintaining Good Standing: You must maintain a grade average of at least a B, earning no more than one grade below B-, and make continuous progress toward graduation. Courses that do not count toward your degree requirements must still meet the B- grade requirement.

Applying for Graduation: Graduation information for graduate students can be found at this website. The chart at the bottom of the page outlines what needs to be done by when for each possible graduation date.
<https://students.tufts.edu/registrar/make-request/apply-graduation/graduation-information-graduate-students>.